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RESOLUTION MSC.539(107) (adopted on 8 June 2023)
2023 AMENDMENTS TO THE
INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
RESOLUTION MSC.539(107)
(adopted on 8 June 2023)
AMENDMENTS TO THE
INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
RECALLING ALSO resolution MSC.268(85) by which it adopted the International Maritime Solid Bulk Cargoes Code (hereinafter referred to as "the IMSBC Code"), which has become mandatory under chapter VI of the International Convention for the Safety of Life at Sea, 1974, as amended ("the Convention"),
RECALLING FURTHER article VIII(b) and regulation VI/1 -1.1 of the Convention concerning amendment procedure for amending the IMSBC Code,
HAVING CONSIDERED, at its 107th session, amendments to the IMSBC Code, proposed
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amendment procedure for amending the IMSBC Code,
HAVING CONSIDERED, at its 107th session, amendments to the IMSBC Code, proposed and circulated in accordance with article VIII(b)(i) of the Convention,
1 ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the IMSBC Code, the text of which is set out in the annex to the present resolution;
2 DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be deemed to have been accepted on 1 July 2024, unless prior to that date, more than one third of the Contracting Governments to the Convention or C ontracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet, have notified their objections to the amendments;
3 INVITES Contracting Governments to the Convention to note that, in accordance with
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3 INVITES Contracting Governments to the Convention to note that, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall enter into force on 1 January 2025 upon their acceptance in accordance with paragraph 2 above;

4 AGREES that Contracting Governments to the Convention may apply the aforementioned amendments in whole or in part on a voluntary basis as from 1 January 2024;

5 REQUESTS the Secretary -General, for the purpose of article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the annex to all Contracting Governments to the Convention; and

6 REQUESTS ALSO the Secretary-General to transmit copies of this resolution and its annex to Members of the Organization, which are not Contracting Governments to the Convention.

Convention.

AMENDMENTS TO THE
INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
AMENDMENT 07-23
(Consolidated version of the IMSBC Code)

The complete text of the IMSBC Code is replaced with the following:

"Table of contents

Section 1 General provisions
Section 2 General loading, carriage and unloading precautions
Section 3 Safety of personnel and ship
Section 4 Assessment of acceptability of consignments for safe shipment
Section 5 Trimming procedures
Section 6 Methods of determining the angle of repose
Section 7 Cargoes which may liquefy or undergo dynamic separation
Section 8 Test procedures for group A cargoes
Section 9 Materials possessing chemical hazards
Section 10 Carriage of solid wastes in bulk
Section 11 Security provisions
Section 12 Stowage factor conversion tables

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Section 10 Carriage of solid wastes in bulk
Section 11 Security provisions
Section 12 Stowage factor conversion tables
Section 13 References to related information and recommendations
Appendix 1 Individual schedules of solid bulk cargoes
Appendix 2 Laboratory test procedures, associated apparatus and standards
Appendix 3 Properties of solid bulk cargoes
Appendix 4 Index
Appendix 5 Bulk Cargo Shipping Names in three languages (English, French and Spanish)

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2023 AMENDMENTS TO THE
INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
Non-c
ommercial use only Non-c
ommercial use only
Sec
tion 1
Gen
eral provisions
1.1 Int
roductory note
It sh
ould be noted that other international and national regulations exist and that those regulations may recognize all or part of the provisions of this Code. In addition, port authorities and other
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may recognize all or part of the provisions of this Code. In addition, port authorities and other bodies and organizations should recognize the Code and may use it as a basis for their storage and handling bye-laws within loading and discharge areas.

1.2 Car
goes listed in this Code

1.2
.1 Typ
ical cargoes currently shipped in bulk, together with advice on their properties and methods of handling, are given in the schedules for individual cargoes. However, these schedules are not exhaustive and the properties attributed to the cargoes are given only for guidance. Consequently, before loading, it is essential to obtain current valid information from the shipper on the physical and chemical properties of the cargoes presented for shipment. The shipper shall provide appropriate information about the cargo to be shipped (see 4.2).

1.2.2 Whe
re a solid bulk cargo is specifically listed in appendix 1 to this Code (individual schedules

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1.2.2 Whe
re a solid bulk cargo is specifically listed in appendix 1 to this Code (individual schedules for solid bulk cargoes), it shall be transported in accordance with the provisions in its schedule in addition to the provisions in sections 1 to 10 and 11.1.1 of this Code. The master shall consider whether to consult the authorities at the ports of loading and discharge, as necessary, concerning the requirements which may be in force and applicable for the carriage.

1.3 Car
goes not listed in this Code

1.3
.1 If a so
lid cargo which is not listed in appendix 1 to this Code is proposed for carriage in bulk, the shipper shall, prior to loading, provide the competent authority of the port of loading with the characteristics and properties of the cargo in accordance with section 4 of this Code. Based on the information received, the competent authority will assess the acceptability of the cargo for safe shipment.

1.3.1.1 Whe

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the information received, the competent authority will assess the acceptability of the cargo for safe shipment.

1.3.1.1 Whe

n it is assessed that the solid bulk cargo proposed for carriage may present hazards as those defined by group A or B of this Code as defined in 1.7 , advice is to be sought from the competent authorities of the port of unloading and of the flag State. The three competent authorities will set the preliminary suitable conditions for the carriage of this cargo.

1.3.1.2 Whe

n it is assessed that the solid bulk cargo proposed for carriage presents no specific hazards for transportation, the carriage of this cargo shall be authorized. The competent authorities of the port of unloading and of the flag State shall be advised of that authorization.

1.3.2 The c

ompetent authority of the port of loading shall provide to the master a certificate stating the characteristics of the cargo and the required conditions for carriage and handling of this shipment.

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characteristics of the cargo and the required conditions for carriage and handling of this shipment.
The competent authority of the port of loading shall also submit an application to the Organization, within one year from the issue of the certificate, to incorporate this solid bulk cargo into appendix 1 of this Code. The format of this application shall be as outlined in 1.3.3.

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1.3
.3 For
mat for the properties of cargoes not listed in this Code
and conditions of the carriage
Ten
tative bulk cargo shipping name (in capital letters)
Description (describe the cargo)
Characteristics (fill the following table)
Phys
ical properties
Size Angle of repose Bulk density (kg/m3) Stow
age factor
(m3/t)
Ha
zard classification
Class Subsidiary hazard(s) MHB Group
Hazard (Clarify the hazard of carriage of the cargo)
(De

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age factor
(m3/t)
Ha
zard classification
Class Subsidiary hazard(s) MHB Group
Hazard (Clarify the hazard of carriage of the cargo)
(De
termine the following types of requirements. If no requirement is necessary, write No special
requirements)
Stowage and segregation
Hold cleanliness
Weather precautions
Loading
Precautions
Ventilation
Carriage
Discharge
Clean-up
(Spe
cify the emergency procedures for the cargo, if necessary)
Emergency procedures
Spe
cial emergency equipment to be carried
Emergency procedures

Emergency action in the event of fire

Medical first aid

Non-c
ommercial use only
1.4 App
lication and implementation of this Code
1.4
.1 The
provisions contained in this Code apply to all ships to which the SOLAS Convention, as
amended, applies and that are carrying solid bulk cargoes as defined in regulation 1-1 of part A of
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amended, applies and that are carrying solid bulk cargoes as defined in regulation 1-1 of part A of chapter VI of the Convention.

1.4.2 Although this Code is legally treated as a mandatory instrument under the SOLAS Convention, the following provisions of this Code remain recommendatory or informative:

Section 11 Security provisions (except 11.1.1);

Section 12 Stowage factor conversion tables;

Section 13 References to related information and recommendations;

Appendices other than appendix 1 (Individual schedules of solid bulk cargoes) and appendix 5 (Bulk Cargo Shipping Names in three languages (English, French and Spanish)); and

The texts in the sections for Description, Characteristics (other than Hazard classification), Hazard and Emergency procedures of individual schedules of solid bulk cargoes in appendix 1.

1.4.3 In certain parts of this Code, a particular action is prescribed, but the responsibility for carrying

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in appendix 1.
1.4.3 In ce
rtain parts of this Code, a particular action is prescribed, but the responsibility for carrying out the action has not been specifically assigned to any particular person. Such responsibility may vary according to the laws and customs of different countries and the international conventions into which these countries have entered. For the purpose of this Code, it is not necessary to make this assignment, but only to identify the action itself. It remains the prerogative of each Government to assign this responsibility.
1.5 Exem
ptions and equivalent measures
1.5
.1 Whe
re this Code requires that a particular provision for the transport of solid bulk cargoes shall be complied with, a competent authority or competent authorities (port State of departure, port State of arrival or flag State) may authorize any other provision by exemption if satisfied that such
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State of arrival or flag State) may authorize any other provision by exemption if satisfied that such provision is at least as effective and safe as that required by this Code. Acceptance of an exemption authorized under this section by a competent authority not party to it is subject to the discretion of that competent authority. Accordingly, prior to any shipment covered by the exemption, the recipient of the exemption shall notify other competent authorities concerned.

1.5.2 A com
petent authority or competent authorities which have taken the initiative with respect to the exemption:

.1 sha
ll send a copy of such exemption to the Organization, which shall bring it to the attention of the Contracting Parties to SOLAS; and

.2 sha
ll take action to amend this Code to include the provisions covered by the exemption, as appropriate.

1.5.3 The p
eriod of validity of the exemption shall be not more than five years from the date of

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appropriate.
1.5.3 The p
eriod of validity of the exemption shall be not more than five years from the date of
authorization. An exemption that is not covered under 1.5.2.2 may be renewed in accordance with
the provisions of this section.
1.5.4 A cop
y of the exemption or an electronic copy thereof shall be maintained on board each ship
transporting solid bulk cargoes in accordance with the exemption, as appropriate.
1.5.5 Con
tact information for the main designated national competent authorities concerned is given in
the separate document issued by the Organization.

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1.6 Conv
entions
Par
ts A and B of chapter VI and part A-1 of chapter VII of the SOLAS Convention, as amended, deal
with the carriage of solid bulk cargoes and the carriage of dangerous goods in solid form in bulk,
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with the carriage of solid bulk cargoes and the carriage of dangerous goods in solid form in bulk, respectively, and only the relevant parts are reproduced below.

Chapt
er VI

Car
riage of cargoes and oil fuels

Part A

Gen
eral provisions

Regu
lation 1

Application

1 Unl
ess expressly provided otherwise, this chapter applies to the carriage of cargoes (except liquids in bulk, gases in bulk and those aspects of carriage covered by other chapters) which, owing to their particular hazards to ships or persons on board, may require special precautions in all ships to which the present regulations apply and in cargo ships of less than 500 gross tonnage. However, for cargo ships of less than 500 gross tonnage, the Administration, if it considers that the sheltered nature and conditions of voyage are such as to render the application of any specific requirements of part A or B of this chapter unreasonable or unnecessary, may take other

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application of any specific requirements of part A or B of this chapter unreasonable or unnecessary, may take other effective measures to ensure the required safety for these ships.

2 To su
pplement the provisions of parts A and B of this chapter, each Contracting Government shall ensure that appropriate information on cargo and its stowage and securing is provided, specifying, in particular, precautions necessary for the safe carriage of such cargoes.

Regu
lation 1-1

Definitions

For t
he purpose of this chapter, unless expressly provided otherwise, the following definitions shall apply:

1 IMS
BC Code means the International Maritime Solid Bulk Cargoes (IMSBC) Code adopted by the Maritime Safety Committee of the Organization by resolution MSC.268(85), as may be amended by the Organization, provided that such amendments are adopted, brought into force and take effect in accordance with the provisions

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provided that such amendments are adopted, brought into force and take effect in accordance with the provisions of article VIII of the present Convention concerning the amendment procedures applicable to the annex other than chapter I.

2 Soli
d bulk cargo means any cargo, other than liquid or gas, consisting of a combination of particles, granules or any larger pieces of material generally uniform in composition, which is loaded directly into the cargo spaces of a ship without any intermediate form of containment.

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ommercial use only

Reg
ulation 1-2

Requirements for the carriage of solid bulk cargoes other than grain

The
carriage of solid bulk cargoes other than grain shall be in compliance with the relevant provisions of the IMSBC Code.

Regu
lation 2

Cargo information

1 The s

IMSBC Code.
Regu
lation 2
Cargo information
1 The s
hipper shall provide the master or their representative with appropriate information on the cargo sufficiently in advance of loading to enable the precautions which may be necessary for proper stowage and safe carriage of the cargo to be put into effect. Such information shall be confirmed in writing and by appropriate shipping documents prior to loading the cargo on the ship.
2 The c
argo information shall include:
.1 in th
e case of general cargo, and of cargo carried in cargo units, a general description of the cargo, the gross mass of the cargo or of the cargo units, and any relevant special properties of the cargo. For the purpose of this regulation the cargo information required in sub-chapter 1.9 of the Code of Safe Practice for Cargo Stowage and Securing, adopted by the Organization by resolution A.714(17), as may be amended,
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for Cargo Stowage and Securing, adopted by the Organization by resolution A.714(17), as may be amended, shall be provided. Any such amendment to sub-chapter 1.9 shall be adopted, brought into force and take effect in accordance with the provisions of article VIII of the present Convention concerning the amendment procedures applicable to the annex other than chapter I;

.2 in the case of solid bulk cargo, information as required by section 4 of the IMSBC Code.

3 Prior to loading cargo units on board ships, the shipper shall ensure that the gross mass of such units is in accordance with the gross mass declared on the shipping documents.

4 In the case of cargo carried in a container, except for containers carried on a chassis or a trailer when such containers are driven on or off a ro-ro ship engaged in short international voyages as defined in regulation III/3, the gross mass according to paragraph 2.1 of this regulation shall be verified by the shipper, either by:

.1 wei

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the gross mass according to paragraph 2.1 of this regulation shall be verified by the shipper, either by:
.1 wei
ghing the packed container using calibrated and certified equipment; or
.2 wei
ghing all packages and cargo items, including the mass of pallets, dunnage and other securing material to be packed in the container and adding the tare mass of the container to the sum of the single masses, using a certified method approved by the competent authority of the State in which packing of the container was completed.
5 The s
hipper of a container shall ensure the verified gross mass is stated in the shipping document. The shipping document shall be:
.1 sig
ned by a person duly authorized by the shipper; and
.2 sub
mitted to the master or their representative and to the terminal representative sufficiently in advance, as required by the master or their representative, to be used in the preparation of the ship stowage plan.
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6 If t
he shipping document, with regard to a packed container, does not provide the verified gross mass and the master or their representative and the terminal representative have not obtained the verified gross mass of the packed container, it shall not be loaded on to the ship.
Regu
lation 3
Oxygen analysis and gas detection equipment
1 Whe
n transporting a solid bulk cargo which is liable to emit a toxic or flammable gas, or cause oxygen depletion in the cargo space, an appropriate instrument for measuring the concentration of gas or oxygen in the air shall be provided together with detailed instructions for its use. Such an instrument shall be to the satisfaction of the Administration.
2 The A
dministration shall take steps to ensure that crews of ships are trained in the use of such instruments.
Regu
lation 4

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the Administration.
2 The A
dministration shall take steps to ensure that crews of ships are trained in the use of such instruments.
Regu
lation 4
The use of pesticides in ships
App
ropriate precautions shall be taken in the use of pesticides in ships, in particular for the purposes of fumigation.
Regu
lation 5
Stowage and securing
1 Car
go, cargo units and cargo transport units carried on or under deck shall be so loaded, stowed and
secured as to prevent as far as is practicable, throughout the voyage, damage or hazard to the ship and the persons
on board, and loss of cargo overboard.
2 Car
go, cargo units and cargo transport units shall be so packed and secured within the unit as to prevent,
throughout the voyage, damage or hazard to the ship and the persons on board.
3 App
ropriate precautions shall be taken during loading and transport of heavy cargoes or cargoes with
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3 App
ropriate precautions shall be taken during loading and transport of heavy cargoes or cargoes with abnormal physical dimensions to ensure that no structural damage to the ship occurs and to maintain adequate stability throughout the voyage.
4 App
ropriate precautions shall be taken during loading and transport of cargo units and cargo transport units on board ro-ro ships, especially with regard to the securing arrangements on board such ships and on the cargo units and cargo transport units and with regard to the strength of the securing points and lashings.
5 Fre
ight containers shall not be loaded to more than the maximum gross weight indicated on the Safety Approval Plate under the International Convention for Safe Containers (CSC), as amended.
6 All c
argoes, other than solid and liquid bulk cargoes, cargo units and cargo transport units, shall be loaded, stowed and secured throughout the voyage in accordance with the Cargo Securing Manual approved by the
10

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stowed and secured throughout the voyage in accordance with the Cargo Securing Manual approved by the
10

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ommercial use only
Adm
inistration. In ships with ro-ro spaces, as defined in regulation II-2/3.41, all securing of such cargoes, cargo units, and cargo transport units, in accordance with the Cargo Securing Manual, shall be completed before the ship leaves the berth. The Cargo Securing Manual shall be drawn up to a standard at least equivalent to relevant guidelines developed by the Organization.
Part
B
Spec
ial provisions for solid bulk cargoes
Regu
lation 6
Acceptability for shipment
1 Pri
or to loading a solid bulk cargo, the master shall be in possession of comprehensive information on the ships stability and on the distribution of cargo for the standard loading conditions. The method of providing such
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ships stability and on the distribution of cargo for the standard loading conditions. The method of providing such information shall be to the satisfaction of the Administration.

Regu
lation 7

Loading, unloading and stowage of solid bulk cargoes

1 For t
he purpose of this regulation, terminal representative means a person appointed by the terminal or other facility, where the ship is loading or unloading, who has responsibility for operations conducted by that terminal or facility with regard to the particular ship.

To enable the master to prevent excessive stresses in the ships structure, the ship shall be provided with a booklet, which shall be written in a language with which the ships officers responsible for cargo operations are familiar. If this language is not English, the ship shall be provided with a booklet written also in the English language. The booklet shall, as a minimum, include:

.1 sta
bility data, as required by regulation II-1/5-1;

.2 bal

---

booklet shall, as a minimum, include:

.1 sta
bility data, as required by regulation II-1/5-1;

.2 bal
lasting and deballasting rates and capacities;

.3 max
imum allowable load per unit surface area of the tank top plating;

.4 max
imum allowable load per hold;

.5 gen
eral loading and unloading instructions with regard to the strength of the ships structure including any limitations on the most adverse operating conditions during loading, unloading, ballasting opera - tions and the voyage;

.6 any s
pecial restrictions such as limitations on the most adverse operating conditions imposed by the Administration or organization recognized by it, if applicable; and

.7 whe
re strength calculations are required, maximum permissible forces and moments on the ships hull during loading, unloading and the voyage.

3 Bef
ore a solid bulk cargo is loaded or unloaded, the master and the terminal representative shall agree on a

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3 Bef
ore a solid bulk cargo is loaded or unloaded, the master and the terminal representative shall agree on a plan which shall ensure that the permissible forces and moments on the ship are not exceeded during loading or unloading, and shall include the sequence, quantity and rate of loading or unloading, taking into consideration the speed of loading or unloading, the number of pours and the deballasting or ballasting capability of the ship. The plan and any subsequent amendments thereto shall be lodged with the appropriate authority of the port State.
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4 The m
aster and terminal representative shall ensure that loading and unloading operations are conducted in accordance with the agreed plan.
5 If du
ring loading or unloading any of the limits of the ship referred to in paragraph 2 are exceeded or are
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accordance with the agreed plan.
5 If du
ring loading or unloading any of the limits of the ship referred to in paragraph 2 are exceeded or are
likely to become so if the loading or unloading continues, the master has the right to suspend operation and the
obligation to notify accordingly the appropriate authority of the port State with which the plan has been lodged.
The master and the terminal representative shall ensure that corrective action is taken. When unloading cargo, the
master and terminal representative shall ensure that the unloading method does not damage the ships structure.
6 The m
aster shall ensure that ships personnel continuously monitor cargo operations. Where possible, the
ships draught shall be checked regularly during loading or unloading to confirm the tonnage figures supplied.
Each draught and tonnage observation shall be recorded in a cargo logbook. If significant deviations from the
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Each draught and tonnage observation shall be recorded in a cargo logbook. If significant deviations from the agreed plan are detected, cargo or ballast operations or both shall be adjusted to ensure that the deviations are corrected.

Chapt
er VII

Car
riage of dangerous goods

Part A-1

Car
riage of dangerous goods in solid form in bulk

Regu
lation 7

Definitions

Dan
gerous goods in solid form in bulk means any material, other than liquid or gas, consisting of a combination of particles, granules or any larger pieces of material, generally uniform in composition, which is covered by the IMDG Code and is loaded directly into the cargo spaces of a ship without any intermediate form of containment, and includes such materials loaded in a barge on a barge-carrying ship.

Regu
lation 7-1

Application

1 Unl
ess expressly provided otherwise, this part applies to the carriage of dangerous goods in solid form in bulk

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Regu
lation 7-1
Application
1 Unl
ess expressly provided otherwise, this part applies to the carriage of dangerous goods in solid form in bulk
in all ships to which the present regulations apply and in cargo ships of less than 500 gross tonnage.
2 The c
arriage of dangerous goods in solid form in bulk is prohibited except in accordance with the provisions
of this part.
3 To su
pplement the provisions of this part, each Contracting Government shall issue, or cause to be issued,
instructions on emergency response and medical first aid relevant to incidents involving dangerous goods in solid
form in bulk, taking into account the guidelines developed by the Organization.

Non-c
ommercial use only
Reg
ulation 7-2
Documents
1 In al
l documents relating to the carriage of dangerous goods in solid form in bulk by sea, the bulk cargo
---

Reg
ulation 7-2
Documents
1 In al
l documents relating to the carriage of dangerous goods in solid form in bulk by sea, the bulk cargo shipping name of the goods shall be used (trade names alone shall not be used).
2 Eac
h ship carrying dangerous goods in solid form in bulk shall have a special list or manifest setting forth the dangerous goods on board and the location thereof. A detailed stowage plan, which identifies by class and sets out the location of all dangerous goods on board, may be used in place of such a special list or manifest. A copy of one of these documents shall be made available before departure to the person or organization designated by the port State authority.
Regu
lation 7-3
Stowage and segregation requirements
1 Dan
gerous goods in solid form in bulk shall be loaded and stowed safely and appropriately in accordance with the nature of the goods. Incompatible goods shall be segregated from one another.
2 Dan

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the nature of the goods. Incompatible goods shall be segregated from one another.
2 Dan
gerous goods in solid form in bulk, which are liable to spontaneous heating or combustion, shall not be carried unless adequate precautions have been taken to minimize the likelihood of the outbreak of fire.
3 Dan
gerous goods in solid form in bulk, which give off dangerous vapours, shall be stowed in a well ventilated cargo space.
Regu
lation 7-4
Reporting of incidents involving dangerous goods
1 Whe
n an incident takes place involving the loss or likely loss overboard of dangerous goods in solid form in bulk into the sea, the master, or other person having charge of the ship, shall report the particulars of such an incident without delay and to the fullest extent possible to the nearest coastal State. The report shall be drawn up based on general principles and guidelines developed by the Organization.
2 In th
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based on general principles and guidelines developed by the Organization.
2 In th
e event of the ship referred to in paragraph 1 being abandoned, or in the event of a report from such a
ship being incomplete or unobtainable, the company, as defined in regulation IX/1.2, shall, to the fullest extent
possible, assume the obligations placed upon the master by this regulation.
Regu
lation 7-5
Requirements for the carriage of dangerous goods in solid form in bulk
The
carriage of dangerous goods in solid form in bulk shall be in compliance with the relevant provisions of the
IMSBC Code, as defined in regulation VI/1-1.1.
1.7 Defin
itions
For t
he purpose of this Code, unless expressly provided otherwise, the following definitions
shall apply:
Angl
e of repose means the maximum slope angle of non-cohesive (i.e. free-flowing) granular
material. It is measured as the angle between a horizontal plane and the cone slope of such material.

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material. It is measured as the angle between a horizontal plane and the cone slope of such material.

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Ang
le of repose
Bul
k Cargo Shipping Name (BCSN) identifies a bulk cargo during transport by sea. When a cargo is listed in this Code, the Bulk Cargo Shipping Name of the cargo is identified by capital letters in the individual schedules or in the index. When a cargo is dangerous goods as defined in the IMDG Code, as defined in regulation VII/1.1 of the SOLAS Convention, refer to 4.1.1.
Bulk d
ensity means the weight of solids, air and water per unit volume. Bulk density is expressed in kilograms per cubic metre (kg/m3), in g
eneral. The void spaces in the cargo may be filled with air
and water.
Car
go space means any space in a ship designated for carriage of cargoes.
Car
---

and water.

Car
go space means any space in a ship designated for carriage of cargoes.

Car
goes which may liquefy means cargoes which contain a certain proportion of fine particles and a certain amount of moisture. They may liquefy if shipped with a moisture content in excess of their transportable moisture limit.

Car
goes which may undergo dynamic separation means cargoes which contain a certain proportion of fine particles and a certain amount of moisture, and may undergo dynamic separation if shipped at a moisture content in excess of their transportable moisture limit.

Coh
esive material means materials other than non-cohesive materials.

Com
petent authority means any national regulatory body or authority designated or otherwise recognized as such for any purpose in connection with this Code. The competent authority shall operate independently from the shipper.

Con
centrates means materials obtained from a natural ore by a process of enrichment or beneficiation
---

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operate independently from the shipper.

Con
centrates means materials obtained from a natural ore by a process of enrichment or beneficiation by physical or chemical separation and removal of unwanted constituents.

Con
signment means a solid bulk cargo presented by a shipper for transport.

Dyna
mic separation means the phenomenon of forming a liquid slurry (water and fine solids) above the solid material, resulting in a free surface effect which may significantly affect the ships stability.

Flo
w moisture point means the percentage moisture content (wet mass basis) at which a flow state develops under the prescribed method of test in a representative sample of the material (see paragraph 1 of appendix 2).

Flow
state means a state occurring when a mass of granular material is saturated with liquid to an extent that, under the influence of prevailing external forces such as vibration, impaction or ships

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extent that, under the influence of prevailing external forces such as vibration, impaction or ships motion, it loses its internal shear strength and behaves as a liquid.

GHS me
ans the ninth revised edition of the Globally Harmonized System of Classification and Labelling of Chemicals, published by the United Nations as document ST/SG/AC.10/30/Rev.9.

Grou
p A consists of cargoes which possess a hazard due to moisture that may result in liquefaction or dynamic separation if shipped at a moisture content in excess of their transportable moisture limit.

Grou
p B consists of cargoes which possess a chemical hazard which could give rise to a dangerous situation on a ship.

Grou
p C consists of cargoes which are classified as neither group A nor group B.

Non-c
ommercial use only
High

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2023 AMENDMENTS TO THE
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Non-c
ommercial use only
High
-density solid bulk cargo means a solid bulk cargo with a stowage factor of 0.56 m3/t or l
ess.
IMDG
Code means the International Maritime Dangerous Goods (IMDG) Code adopted by the
Maritime Safety Committee of the Organization by resolution MSC.122(75), as may be amended by
the Organization.
Inc
ompatible materials means materials that may react dangerously when mixed. They are subject to
the segregation requirements of 9.3 and the schedules for individual cargoes classified in group B.
Int
ernational Ship and Port Facility Security (ISPS) Code means the International Code for the
Security of Ships and of Port Facilities, consisting of Part A (the provisions of which shall be treated
as mandatory) and Part B (the provisions of which shall be treated as recommendatory), as adopted,
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as mandatory) and Part B (the provisions of which shall be treated as recommendatory), as adopted, on 12 December 2002, by resolution 2 of the Conference of Contracting Governments to the International Convention for the Safety of Life at Sea, 1974, as may be amended by the Organization.

Manual of Tests and Criteria means the seventh revised edition of the United Nations publication entitled Manual of Tests and Criteria (ST/SG/AC.10/11/Rev.7 and Amendment 1).

Materials hazardous only in bulk (MHB) means materials which may possess chemical hazards when carried in bulk other than materials classified as dangerous goods in the IMDG Code.

Moisture content means that portion of a representative sample consisting of water, ice or other liquid expressed as a percentage of the total wet mass of that sample.

Moisture migration means the movement of moisture contained in a cargo by settling and

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Moi
sture migration means the movement of moisture contained in a cargo by settling and consolidation of the cargo due to vibration and ships motion. Water is progressively displaced, which may result in some portions or all of the cargo developing a flow state.

Non
-cohesive material means dry materials that readily shift due to sliding during transport, as listed in appendix 3, paragraph 1, Properties of dry bulk cargoes.

Pot
ential sources of ignition means, but is not limited to, open fires, machinery exhausts, galley uptakes, electrical outlets and electrical equipment unless they are of certified safe type.

Rep
resentative test sample means a sample of sufficient quantity for the purpose of testing the physical and chemical properties of the consignment to meet specified requirements.

Ship
per means any person by whom or in whose name, or on whose behalf, a contract of carriage

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Ship
per means any person by whom or in whose name, or on whose behalf, a contract of carriage of goods by sea has been concluded with a carrier, or any person by whom or in whose name, or on whose behalf, the goods are actually delivered to the carrier in relation to the contract of carriage by sea.

Sol
id bulk cargo means any cargo, other than a liquid or a gas, consisting of a combination of particles, granules or any larger pieces of material generally uniform in composition which is loaded directly into the cargo spaces of a ship without any intermediate form of containment.

Sour
ces of heat means heated ship structures, where the surface temperature is liable to exceed 55ºC. Examples of such heated structures are steam pipes, heating coils, top or side walls of heated fuel and cargo tanks, and bulkheads of machinery spaces.

Stow
age factor means the figure which expresses the number of cubic metres which one tonne of cargo will occupy.

Tra

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Stow
age factor means the figure which expresses the number of cubic metres which one tonne of cargo will occupy.
Tra
nsportable moisture limit (TML) of a group A cargo means the maximum moisture content of the cargo which is considered safe for carriage in ships not complying with the special provisions of 7 .3.2. It is determined by the test procedures, approved by a competent authority, such as those specified in paragraph 1 of appendix 2.
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Trim
ming means any levelling of a cargo within a cargo space, either partial or total.
Vent
ilation means exchange of air from outside to inside a cargo space.
.1 Cont
inuous ventilation means ventilation that is operating at all times.
.2 Mech
anical ventilation means power-generated ventilation.
.3 Nat
ural ventilation means ventilation that is not power-generated.
.4 Sur

.2 Mech
anical ventilation means power-generated ventilation.
.3 Nat
ural ventilation means ventilation that is not power-generated.
.4 Sur
face ventilation means ventilation of the space above the cargo.
16
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RESOLUTION MSC.539(107)
(adopted on 8 June 2023)
AMENDMENTS TO THE
INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
RECALLING ALSO resolution MSC.268(85) by which it adopted the International Maritime Solid Bulk Cargoes Code (hereinafter referred to as "the IMSBC Code"), which has become mandatory under chapter VI of the International Convention for the Safety of Life at Sea, 1974, as amended ("the Convention"),
RECALLING FURTHER article VIII(b) and regulation VI/1 -1.1 of the Convention concerning amendment procedure for amending the IMSBC Code,
HAVING CONSIDERED, at its 107th session, amendments to the IMSBC Code, proposed
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amendment procedure for amending the IMSBC Code,
HAVING CONSIDERED, at its 107th session, amendments to the IMSBC Code, proposed and circulated in accordance with article VIII(b)(i) of the Convention,
1 ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the IMSBC Code, the text of which is set out in the annex to the present resolution;
2 DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be deemed to have been accepted on 1 July 2024, unless prior to that date, more than one third of the Contracting Governments to the Convention or C ontracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet, have notified their objections to the amendments;
3 INVITES Contracting Governments to the Convention to note that, in accordance with
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3 INVITES Contracting Governments to the Convention to note that, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall enter into force on 1 January 2025 upon their acceptance in accordance with paragraph 2 above;

4 AGREES that Contracting Governments to the Convention may apply the aforementioned amendments in whole or in part on a voluntary basis as from 1 January 2024;

5 REQUESTS the Secretary -General, for the purpose of article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the annex to all Contracting Governments to the Convention; and

6 REQUESTS ALSO the Secretary-General to transmit copies of this resolution and its annex to Members of the Organization, which are not Contracting Governments to the Convention.

3

RESOLUTION MSC.539(107) (adopted on 8 June 2023)

2023 AMENDMENTS TO THE

INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE

Convention.

RESOLUTION MSC.539(107) (adopted on 8 June 2023)
2023 AMENDMENTS TO THE
INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
ANNEX

AMENDMENTS TO THE
INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
AMENDMENT 07-23
(Consolidated version of the IMSBC Code)

The complete text of the IMSBC Code is replaced with the following:

"Table of contents

Section 1 General provisions
Section 2 General loading, carriage and unloading precautions
Section 3 Safety of personnel and ship
Section 4 Assessment of acceptability of consignments for safe shipment
Section 5 Trimming procedures
Section 6 Methods of determining the angle of repose
Section 7 Cargoes which may liquefy or undergo dynamic separation
Section 8 Test procedures for group A cargoes
Section 9 Materials possessing chemical hazards
Section 10 Carriage of solid wastes in bulk
Section 11 Security provisions
Section 12 Stowage factor conversion tables

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Section 10 Carriage of solid wastes in bulk
Section 11 Security provisions
Section 12 Stowage factor conversion tables
Section 13 References to related information and recommendations
Appendix 1 Individual schedules of solid bulk cargoes
Appendix 2 Laboratory test procedures, associated apparatus and standards
Appendix 3 Properties of solid bulk cargoes
Appendix 4 Index
Appendix 5 Bulk Cargo Shipping Names in three languages (English, French and Spanish)

Non-c
ommercial use only Non-c
ommercial use only
Sec
tion 1
Gen
eral provisions
1.1 Int
roductory note
It sh
ould be noted that other international and national regulations exist and that those regulations may recognize all or part of the provisions of this Code. In addition, port authorities and other
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may recognize all or part of the provisions of this Code. In addition, port authorities and other bodies and organizations should recognize the Code and may use it as a basis for their storage and handling bye-laws within loading and discharge areas.

1.2 Car
goes listed in this Code

1.2
.1 Typ
ical cargoes currently shipped in bulk, together with advice on their properties and methods of handling, are given in the schedules for individual cargoes. However, these schedules are not exhaustive and the properties attributed to the cargoes are given only for guidance. Consequently, before loading, it is essential to obtain current valid information from the shipper on the physical and chemical properties of the cargoes presented for shipment. The shipper shall provide appropriate information about the cargo to be shipped (see 4.2).

1.2.2 Whe
re a solid bulk cargo is specifically listed in appendix 1 to this Code (individual schedules

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1.2.2 Whe
re a solid bulk cargo is specifically listed in appendix 1 to this Code (individual schedules for solid bulk cargoes), it shall be transported in accordance with the provisions in its schedule in addition to the provisions in sections 1 to 10 and 11.1.1 of this Code. The master shall consider whether to consult the authorities at the ports of loading and discharge, as necessary, concerning the requirements which may be in force and applicable for the carriage.

1.3 Car
goes not listed in this Code

1.3
.1 If a so
lid cargo which is not listed in appendix 1 to this Code is proposed for carriage in bulk, the shipper shall, prior to loading, provide the competent authority of the port of loading with the characteristics and properties of the cargo in accordance with section 4 of this Code. Based on the information received, the competent authority will assess the acceptability of the cargo for safe shipment.

1.3.1.1 Whe

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the information received, the competent authority will assess the acceptability of the cargo for safe shipment.

1.3.1.1 Whe

n it is assessed that the solid bulk cargo proposed for carriage may present hazards as those defined by group A or B of this Code as defined in 1.7 , advice is to be sought from the competent authorities of the port of unloading and of the flag State. The three competent authorities will set the preliminary suitable conditions for the carriage of this cargo.

1.3.1.2 Whe

n it is assessed that the solid bulk cargo proposed for carriage presents no specific hazards for transportation, the carriage of this cargo shall be authorized. The competent authorities of the port of unloading and of the flag State shall be advised of that authorization.

1.3.2 The c

ompetent authority of the port of loading shall provide to the master a certificate stating the characteristics of the cargo and the required conditions for carriage and handling of this shipment.

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characteristics of the cargo and the required conditions for carriage and handling of this shipment.
The competent authority of the port of loading shall also submit an application to the Organization, within one year from the issue of the certificate, to incorporate this solid bulk cargo into appendix 1 of this Code. The format of this application shall be as outlined in 1.3.3.
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1.3
.3 For
mat for the properties of cargoes not listed in this Code
and conditions of the carriage
Ten
tative bulk cargo shipping name (in capital letters)
Description (describe the cargo)
Characteristics (fill the following table)
Phys
ical properties
Size Angle of repose Bulk density (kg/m3) Stow
age factor
(m3/t)
Ha
zard classification
Class Subsidiary hazard(s) MHB Group
Hazard (Clarify the hazard of carriage of the cargo)
(De

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age factor
(m3/t)
Ha
zard classification
Class Subsidiary hazard(s) MHB Group
Hazard (Clarify the hazard of carriage of the cargo)
(De
termine the following types of requirements. If no requirement is necessary, write No special
requirements)
Stowage and segregation
Hold cleanliness
Weather precautions
Loading
Precautions
Ventilation
Carriage
Discharge
Clean-up
(Spe
cify the emergency procedures for the cargo, if necessary)
Emergency procedures
Spe
cial emergency equipment to be carried
Emergency procedures

Emergency action in the event of fire

Medical first aid

Non-c
ommercial use only
1.4 App
lication and implementation of this Code
1.4
.1 The
provisions contained in this Code apply to all ships to which the SOLAS Convention, as
amended, applies and that are carrying solid bulk cargoes as defined in regulation 1-1 of part A of
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amended, applies and that are carrying solid bulk cargoes as defined in regulation 1-1 of part A of chapter VI of the Convention.

1.4.2 Although this Code is legally treated as a mandatory instrument under the SOLAS Convention, the following provisions of this Code remain recommendatory or informative:

Section 11 Security provisions (except 11.1.1);

Section 12 Stowage factor conversion tables;

Section 13 References to related information and recommendations;

Appendices other than appendix 1 (Individual schedules of solid bulk cargoes) and appendix 5 (Bulk Cargo Shipping Names in three languages (English, French and Spanish)); and

The texts in the sections for Description, Characteristics (other than Hazard classification), Hazard and Emergency procedures of individual schedules of solid bulk cargoes in appendix 1.

1.4.3 In certain parts of this Code, a particular action is prescribed, but the responsibility for carrying

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in appendix 1.

1.4.3 In ce
rtain parts of this Code, a particular action is prescribed, but the responsibility for carrying out the action has not been specifically assigned to any particular person. Such responsibility may vary according to the laws and customs of different countries and the international conventions into which these countries have entered. For the purpose of this Code, it is not necessary to make this assignment, but only to identify the action itself. It remains the prerogative of each Government to assign this responsibility.

1.5 Exem
ptions and equivalent measures

1.5
.1 Whe
re this Code requires that a particular provision for the transport of solid bulk cargoes shall be complied with, a competent authority or competent authorities (port State of departure, port State of arrival or flag State) may authorize any other provision by exemption if satisfied that such

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State of arrival or flag State) may authorize any other provision by exemption if satisfied that such provision is at least as effective and safe as that required by this Code. Acceptance of an exemption authorized under this section by a competent authority not party to it is subject to the discretion of that competent authority. Accordingly, prior to any shipment covered by the exemption, the recipient of the exemption shall notify other competent authorities concerned.

1.5.2 A com
petent authority or competent authorities which have taken the initiative with respect to the exemption:

.1 sha
ll send a copy of such exemption to the Organization, which shall bring it to the attention of the Contracting Parties to SOLAS; and

.2 sha
ll take action to amend this Code to include the provisions covered by the exemption, as appropriate.

1.5.3 The p
eriod of validity of the exemption shall be not more than five years from the date of

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appropriate.
1.5.3 The p
eriod of validity of the exemption shall be not more than five years from the date of
authorization. An exemption that is not covered under 1.5.2.2 may be renewed in accordance with
the provisions of this section.
1.5.4 A cop
y of the exemption or an electronic copy thereof shall be maintained on board each ship
transporting solid bulk cargoes in accordance with the exemption, as appropriate.
1.5.5 Con
tact information for the main designated national competent authorities concerned is given in
the separate document issued by the Organization.

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1.6 Conv
entions
Par
ts A and B of chapter VI and part A-1 of chapter VII of the SOLAS Convention, as amended, deal
with the carriage of solid bulk cargoes and the carriage of dangerous goods in solid form in bulk,
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with the carriage of solid bulk cargoes and the carriage of dangerous goods in solid form in bulk, respectively, and only the relevant parts are reproduced below.

Chapt
er VI

Car
riage of cargoes and oil fuels

Part A

Gen
eral provisions

Regu
lation 1

Application

1 Unl
ess expressly provided otherwise, this chapter applies to the carriage of cargoes (except liquids in bulk, gases in bulk and those aspects of carriage covered by other chapters) which, owing to their particular hazards to ships or persons on board, may require special precautions in all ships to which the present regulations apply and in cargo ships of less than 500 gross tonnage. However, for cargo ships of less than 500 gross tonnage, the Administration, if it considers that the sheltered nature and conditions of voyage are such as to render the application of any specific requirements of part A or B of this chapter unreasonable or unnecessary, may take other

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application of any specific requirements of part A or B of this chapter unreasonable or unnecessary, may take other effective measures to ensure the required safety for these ships.

2 To su
pplement the provisions of parts A and B of this chapter, each Contracting Government shall ensure that appropriate information on cargo and its stowage and securing is provided, specifying, in particular, precautions necessary for the safe carriage of such cargoes.

Regu
lation 1-1

Definitions

For t
he purpose of this chapter, unless expressly provided otherwise, the following definitions shall apply:

1 IMS
BC Code means the International Maritime Solid Bulk Cargoes (IMSBC) Code adopted by the Maritime Safety Committee of the Organization by resolution MSC.268(85), as may be amended by the Organization, provided that such amendments are adopted, brought into force and take effect in accordance with the provisions

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provided that such amendments are adopted, brought into force and take effect in accordance with the provisions of article VIII of the present Convention concerning the amendment procedures applicable to the annex other than chapter I.

2 Soli
d bulk cargo means any cargo, other than liquid or gas, consisting of a combination of particles, granules or any larger pieces of material generally uniform in composition, which is loaded directly into the cargo spaces of a ship without any intermediate form of containment.

Non-c
ommercial use only

Reg
ulation 1-2
Requirements for the carriage of solid bulk cargoes other than grain
The
carriage of solid bulk cargoes other than grain shall be in compliance with the relevant provisions of the IMSBC Code.

Regu
lation 2
Cargo information
1 The s

IMSBC Code.
Regu
lation 2
Cargo information
1 The s
hipper shall provide the master or their representative with appropriate information on the cargo sufficiently in advance of loading to enable the precautions which may be necessary for proper stowage and safe carriage of the cargo to be put into effect. Such information shall be confirmed in writing and by appropriate shipping documents prior to loading the cargo on the ship.
2 The c
argo information shall include:
.1 in th
e case of general cargo, and of cargo carried in cargo units, a general description of the cargo, the gross mass of the cargo or of the cargo units, and any relevant special properties of the cargo. For the purpose of this regulation the cargo information required in sub-chapter 1.9 of the Code of Safe Practice for Cargo Stowage and Securing, adopted by the Organization by resolution A.714(17), as may be amended,
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for Cargo Stowage and Securing, adopted by the Organization by resolution A.714(17), as may be amended, shall be provided. Any such amendment to sub-chapter 1.9 shall be adopted, brought into force and take effect in accordance with the provisions of article VIII of the present Convention concerning the amendment procedures applicable to the annex other than chapter I;

.2 in th
e case of solid bulk cargo, information as required by section 4 of the IMSBC Code.

3 Pri
or to loading cargo units on board ships, the shipper shall ensure that the gross mass of such units is in accordance with the gross mass declared on the shipping documents.

4 In th
e case of cargo carried in a container, except for containers carried on a chassis or a trailer when such containers are driven on or off a ro-ro ship engaged in short international voyages as defined in regulation III/3, the gross mass according to paragraph 2.1 of this regulation shall be verified by the shipper, either by:

.1 wei

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the gross mass according to paragraph 2.1 of this regulation shall be verified by the shipper, either by:
.1 wei
ghing the packed container using calibrated and certified equipment; or
.2 wei
ghing all packages and cargo items, including the mass of pallets, dunnage and other securing material to be packed in the container and adding the tare mass of the container to the sum of the single masses, using a certified method approved by the competent authority of the State in which packing of the container was completed.
5 The s
hipper of a container shall ensure the verified gross mass is stated in the shipping document. The shipping document shall be:
.1 sig
ned by a person duly authorized by the shipper; and
.2 sub
mitted to the master or their representative and to the terminal representative sufficiently in advance, as required by the master or their representative, to be used in the preparation of the ship stowage plan.
9
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6 If t
he shipping document, with regard to a packed container, does not provide the verified gross mass and the master or their representative and the terminal representative have not obtained the verified gross mass of the packed container, it shall not be loaded on to the ship.
Regu
lation 3
Oxygen analysis and gas detection equipment
1 Whe
n transporting a solid bulk cargo which is liable to emit a toxic or flammable gas, or cause oxygen depletion in the cargo space, an appropriate instrument for measuring the concentration of gas or oxygen in the air shall be provided together with detailed instructions for its use. Such an instrument shall be to the satisfaction of the Administration.
2 The A
dministration shall take steps to ensure that crews of ships are trained in the use of such instruments.
Regu
lation 4

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the Administration.
2 The A
dministration shall take steps to ensure that crews of ships are trained in the use of such instruments.
Regu
lation 4
The use of pesticides in ships
App
ropriate precautions shall be taken in the use of pesticides in ships, in particular for the purposes of fumigation.
Regu
lation 5
Stowage and securing
1 Car
go, cargo units and cargo transport units carried on or under deck shall be so loaded, stowed and
secured as to prevent as far as is practicable, throughout the voyage, damage or hazard to the ship and the persons
on board, and loss of cargo overboard.
2 Car
go, cargo units and cargo transport units shall be so packed and secured within the unit as to prevent,
throughout the voyage, damage or hazard to the ship and the persons on board.
3 App
ropriate precautions shall be taken during loading and transport of heavy cargoes or cargoes with
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3 App
ropriate precautions shall be taken during loading and transport of heavy cargoes or cargoes with abnormal physical dimensions to ensure that no structural damage to the ship occurs and to maintain adequate stability throughout the voyage.
4 App
ropriate precautions shall be taken during loading and transport of cargo units and cargo transport units on board ro-ro ships, especially with regard to the securing arrangements on board such ships and on the cargo units and cargo transport units and with regard to the strength of the securing points and lashings.
5 Fre
ight containers shall not be loaded to more than the maximum gross weight indicated on the Safety Approval Plate under the International Convention for Safe Containers (CSC), as amended.
6 All c
argoes, other than solid and liquid bulk cargoes, cargo units and cargo transport units, shall be loaded, stowed and secured throughout the voyage in accordance with the Cargo Securing Manual approved by the
10

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stowed and secured throughout the voyage in accordance with the Cargo Securing Manual approved by the
10

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ommercial use only
Adm
inistration. In ships with ro-ro spaces, as defined in regulation II-2/3.41, all securing of such cargoes, cargo units, and cargo transport units, in accordance with the Cargo Securing Manual, shall be completed before the ship leaves the berth. The Cargo Securing Manual shall be drawn up to a standard at least equivalent to relevant guidelines developed by the Organization.
Part
B
Spec
ial provisions for solid bulk cargoes
Regu
lation 6
Acceptability for shipment
1 Pri
or to loading a solid bulk cargo, the master shall be in possession of comprehensive information on the ships stability and on the distribution of cargo for the standard loading conditions. The method of providing such
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ships stability and on the distribution of cargo for the standard loading conditions. The method of providing such information shall be to the satisfaction of the Administration.

Regu
lation 7

Loading, unloading and stowage of solid bulk cargoes

1 For t
he purpose of this regulation, terminal representative means a person appointed by the terminal or other facility, where the ship is loading or unloading, who has responsibility for operations conducted by that terminal or facility with regard to the particular ship.

To enable the master to prevent excessive stresses in the ships structure, the ship shall be provided with a booklet, which shall be written in a language with which the ships officers responsible for cargo operations are familiar. If this language is not English, the ship shall be provided with a booklet written also in the English language. The booklet shall, as a minimum, include:

.1 sta
bility data, as required by regulation II-1/5-1;

.2 bal

---

booklet shall, as a minimum, include:
.1 sta
bility data, as required by regulation II-1/5-1;
.2 bal
lasting and deballasting rates and capacities;
.3 max
imum allowable load per unit surface area of the tank top plating;
.4 max
imum allowable load per hold;
.5 gen
eral loading and unloading instructions with regard to the strength of the ships structure including any limitations on the most adverse operating conditions during loading, unloading, ballasting opera - tions and the voyage;
.6 any s
pecial restrictions such as limitations on the most adverse operating conditions imposed by the Administration or organization recognized by it, if applicable; and
.7 whe
re strength calculations are required, maximum permissible forces and moments on the ships hull during loading, unloading and the voyage.
3 Bef
ore a solid bulk cargo is loaded or unloaded, the master and the terminal representative shall agree on a
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3 Bef
ore a solid bulk cargo is loaded or unloaded, the master and the terminal representative shall agree on a plan which shall ensure that the permissible forces and moments on the ship are not exceeded during loading or unloading, and shall include the sequence, quantity and rate of loading or unloading, taking into consideration the speed of loading or unloading, the number of pours and the deballasting or ballasting capability of the ship. The plan and any subsequent amendments thereto shall be lodged with the appropriate authority of the port State.
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4 The m
aster and terminal representative shall ensure that loading and unloading operations are conducted in accordance with the agreed plan.
5 If du
ring loading or unloading any of the limits of the ship referred to in paragraph 2 are exceeded or are
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accordance with the agreed plan.

5 If du
ring loading or unloading any of the limits of the ship referred to in paragraph 2 are exceeded or are likely to become so if the loading or unloading continues, the master has the right to suspend operation and the obligation to notify accordingly the appropriate authority of the port State with which the plan has been lodged. The master and the terminal representative shall ensure that corrective action is taken. When unloading cargo, the master and terminal representative shall ensure that the unloading method does not damage the ships structure.

6 The m
aster shall ensure that ships personnel continuously monitor cargo operations. Where possible, the ships draught shall be checked regularly during loading or unloading to confirm the tonnage figures supplied. Each draught and tonnage observation shall be recorded in a cargo logbook. If significant deviations from the

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Each draught and tonnage observation shall be recorded in a cargo logbook. If significant deviations from the agreed plan are detected, cargo or ballast operations or both shall be adjusted to ensure that the deviations are corrected.

Chapt
er VII

Car
riage of dangerous goods

Part A-1

Car
riage of dangerous goods in solid form in bulk

Regu
lation 7

Definitions

Dan
gerous goods in solid form in bulk means any material, other than liquid or gas, consisting of a combination of particles, granules or any larger pieces of material, generally uniform in composition, which is covered by the IMDG Code and is loaded directly into the cargo spaces of a ship without any intermediate form of containment, and includes such materials loaded in a barge on a barge-carrying ship.

Regu
lation 7-1

Application

1 Unl
ess expressly provided otherwise, this part applies to the carriage of dangerous goods in solid form in bulk

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Regu
lation 7-1
Application
1 Unl
ess expressly provided otherwise, this part applies to the carriage of dangerous goods in solid form in bulk
in all ships to which the present regulations apply and in cargo ships of less than 500 gross tonnage.
2 The c
arriage of dangerous goods in solid form in bulk is prohibited except in accordance with the provisions
of this part.
3 To su
pplement the provisions of this part, each Contracting Government shall issue, or cause to be issued,
instructions on emergency response and medical first aid relevant to incidents involving dangerous goods in solid
form in bulk, taking into account the guidelines developed by the Organization.

Non-c
ommercial use only
Reg
ulation 7-2
Documents
1 In al
l documents relating to the carriage of dangerous goods in solid form in bulk by sea, the bulk cargo
---

Reg
ulation 7-2
Documents
1 In al
l documents relating to the carriage of dangerous goods in solid form in bulk by sea, the bulk cargo shipping name of the goods shall be used (trade names alone shall not be used).
2 Eac
h ship carrying dangerous goods in solid form in bulk shall have a special list or manifest setting forth the dangerous goods on board and the location thereof. A detailed stowage plan, which identifies by class and sets out the location of all dangerous goods on board, may be used in place of such a special list or manifest. A copy of one of these documents shall be made available before departure to the person or organization designated by the port State authority.
Regu
lation 7-3
Stowage and segregation requirements
1 Dan
gerous goods in solid form in bulk shall be loaded and stowed safely and appropriately in accordance with the nature of the goods. Incompatible goods shall be segregated from one another.
2 Dan

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the nature of the goods. Incompatible goods shall be segregated from one another.
2 Dan
gerous goods in solid form in bulk, which are liable to spontaneous heating or combustion, shall not be carried unless adequate precautions have been taken to minimize the likelihood of the outbreak of fire.
3 Dan
gerous goods in solid form in bulk, which give off dangerous vapours, shall be stowed in a well ventilated cargo space.
Regu
lation 7-4
Reporting of incidents involving dangerous goods
1 Whe
n an incident takes place involving the loss or likely loss overboard of dangerous goods in solid form in bulk into the sea, the master, or other person having charge of the ship, shall report the particulars of such an incident without delay and to the fullest extent possible to the nearest coastal State. The report shall be drawn up based on general principles and guidelines developed by the Organization.
2 In th

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based on general principles and guidelines developed by the Organization.

2 In th
e event of the ship referred to in paragraph 1 being abandoned, or in the event of a report from such a ship being incomplete or unobtainable, the company, as defined in regulation IX/1.2, shall, to the fullest extent possible, assume the obligations placed upon the master by this regulation.

Regu
lation 7-5

Requirements for the carriage of dangerous goods in solid form in bulk

The
carriage of dangerous goods in solid form in bulk shall be in compliance with the relevant provisions of the IMSBC Code, as defined in regulation VI/1-1.1.

1.7 Defin
itions

For t
he purpose of this Code, unless expressly provided otherwise, the following definitions shall apply:

Angl
e of repose means the maximum slope angle of non-cohesive (i.e. free-flowing) granular material. It is measured as the angle between a horizontal plane and the cone slope of such material.

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material. It is measured as the angle between a horizontal plane and the cone slope of such material.

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Ang
le of repose
Bul
k Cargo Shipping Name (BCSN) identifies a bulk cargo during transport by sea. When a cargo is listed in this Code, the Bulk Cargo Shipping Name of the cargo is identified by capital letters in the individual schedules or in the index. When a cargo is dangerous goods as defined in the IMDG Code, as defined in regulation VII/1.1 of the SOLAS Convention, refer to 4.1.1.
Bulk d
ensity means the weight of solids, air and water per unit volume. Bulk density is expressed in kilograms per cubic metre (kg/m3), in g
eneral. The void spaces in the cargo may be filled with air
and water.
Car
go space means any space in a ship designated for carriage of cargoes.
Car
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and water.

Car
go space means any space in a ship designated for carriage of cargoes.

Car
goes which may liquefy means cargoes which contain a certain proportion of fine particles and a certain amount of moisture. They may liquefy if shipped with a moisture content in excess of their transportable moisture limit.

Car
goes which may undergo dynamic separation means cargoes which contain a certain proportion of fine particles and a certain amount of moisture, and may undergo dynamic separation if shipped at a moisture content in excess of their transportable moisture limit.

Coh
esive material means materials other than non-cohesive materials.

Com
petent authority means any national regulatory body or authority designated or otherwise recognized as such for any purpose in connection with this Code. The competent authority shall operate independently from the shipper.

Con
centrates means materials obtained from a natural ore by a process of enrichment or beneficiation

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operate independently from the shipper.
Con
centrates means materials obtained from a natural ore by a process of enrichment or beneficiation by physical or chemical separation and removal of unwanted constituents.
Con
signment means a solid bulk cargo presented by a shipper for transport.
Dyna
mic separation means the phenomenon of forming a liquid slurry (water and fine solids) above the solid material, resulting in a free surface effect which may significantly affect the ships stability.
Flo
w moisture point means the percentage moisture content (wet mass basis) at which a flow state develops under the prescribed method of test in a representative sample of the material (see paragraph 1 of appendix 2).
Flow
state means a state occurring when a mass of granular material is saturated with liquid to an extent that, under the influence of prevailing external forces such as vibration, impaction or ships
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extent that, under the influence of prevailing external forces such as vibration, impaction or ships motion, it loses its internal shear strength and behaves as a liquid.

GHS means the ninth revised edition of the Globally Harmonized System of Classification and Labelling of Chemicals, published by the United Nations as document ST/SG/AC.10/30/Rev.9.

Group A consists of cargoes which possess a hazard due to moisture that may result in liquefaction or dynamic separation if shipped at a moisture content in excess of their transportable moisture limit.

Group B consists of cargoes which possess a chemical hazard which could give rise to a dangerous situation on a ship.

Group C consists of cargoes which are classified as neither group A nor group B.

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High

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Non-c
ommercial use only
High
-density solid bulk cargo means a solid bulk cargo with a stowage factor of 0.56 m3/t or l
ess.
IMDG
Code means the International Maritime Dangerous Goods (IMDG) Code adopted by the
Maritime Safety Committee of the Organization by resolution MSC.122(75), as may be amended by
the Organization.
Inc
ompatible materials means materials that may react dangerously when mixed. They are subject to
the segregation requirements of 9.3 and the schedules for individual cargoes classified in group B.
Int
ernational Ship and Port Facility Security (ISPS) Code means the International Code for the
Security of Ships and of Port Facilities, consisting of Part A (the provisions of which shall be treated
as mandatory) and Part B (the provisions of which shall be treated as recommendatory), as adopted,
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as mandatory) and Part B (the provisions of which shall be treated as recommendatory), as adopted, on 12 December 2002, by resolution 2 of the Conference of Contracting Governments to the International Convention for the Safety of Life at Sea, 1974, as may be amended by the Organization.

Manual of Tests and Criteria means the seventh revised edition of the United Nations publication entitled Manual of Tests and Criteria (ST/SG/AC.10/11/Rev.7 and Amendment 1).

Materials hazardous only in bulk (MHB) means materials which may possess chemical hazards when carried in bulk other than materials classified as dangerous goods in the IMDG Code.

Moisture content means that portion of a representative sample consisting of water, ice or other liquid expressed as a percentage of the total wet mass of that sample.

Moisture migration means the movement of moisture contained in a cargo by settling and

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Moi
sture migration means the movement of moisture contained in a cargo by settling and consolidation of the cargo due to vibration and ships motion. Water is progressively displaced, which may result in some portions or all of the cargo developing a flow state.

Non
-cohesive material means dry materials that readily shift due to sliding during transport, as listed in appendix 3, paragraph 1, Properties of dry bulk cargoes.

Pot
ential sources of ignition means, but is not limited to, open fires, machinery exhausts, galley uptakes, electrical outlets and electrical equipment unless they are of certified safe type.

Rep
resentative test sample means a sample of sufficient quantity for the purpose of testing the physical and chemical properties of the consignment to meet specified requirements.

Ship
per means any person by whom or in whose name, or on whose behalf, a contract of carriage

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Ship
per means any person by whom or in whose name, or on whose behalf, a contract of carriage of goods by sea has been concluded with a carrier, or any person by whom or in whose name, or on whose behalf, the goods are actually delivered to the carrier in relation to the contract of carriage by sea.

Sol
id bulk cargo means any cargo, other than a liquid or a gas, consisting of a combination of particles, granules or any larger pieces of material generally uniform in composition which is loaded directly into the cargo spaces of a ship without any intermediate form of containment.

Sour
ces of heat means heated ship structures, where the surface temperature is liable to exceed 55ºC. Examples of such heated structures are steam pipes, heating coils, top or side walls of heated fuel and cargo tanks, and bulkheads of machinery spaces.

Stow
age factor means the figure which expresses the number of cubic metres which one tonne of cargo will occupy.

Tra

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Stow
age factor means the figure which expresses the number of cubic metres which one tonne of cargo will occupy.
Tra
nsportable moisture limit (TML) of a group A cargo means the maximum moisture content of the cargo which is considered safe for carriage in ships not complying with the special provisions of 7 .3.2. It is determined by the test procedures, approved by a competent authority, such as those specified in paragraph 1 of appendix 2.

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Trim
ming means any levelling of a cargo within a cargo space, either partial or total.
Vent
ilation means exchange of air from outside to inside a cargo space.
.1 Cont
inuous ventilation means ventilation that is operating at all times.
.2 Mech
anical ventilation means power-generated ventilation.
.3 Nat
ural ventilation means ventilation that is not power-generated.
.4 Sur

.2 Mech
anical ventilation means power-generated ventilation.
.3 Nat
ural ventilation means ventilation that is not power-generated.
.4 Sur
face ventilation means ventilation of the space above the cargo.
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